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The continuous-coefficient Jensen equation: A note on vestigial regularity hypotheses

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The continuous-coefficient Jensen equation: A note on vestigial regularity hypotheses

Double Blind No Name

Abstract. Jensen's inequality is a workhorse of analysis: for every concave function G and random variable ξ on an interval, $\mathbb{E}[G(\xi)] \leq G(\mathbb{E}[\xi])$. What if equality is forced for every two-point distribution — i.e., $pG(u_1) + (1-p)G(u_2) = G(pu_1 + (1-p)u_2)$ for all u_1, u_2 in the interval and all weights $p \in [0, 1]$? Must G be affine?

Yes. The proof is one line: substitute $u_1 = M$, $u_2 = 0$, $p = v/M$ and read $G(v)$ off the straight line through $(0, G(0))$ and $(M, G(M))$. No regularity hypothesis on G — continuity, measurability, monotonicity, or boundedness — is needed.

The result contrasts sharply with what happens when the weights p are restricted to rationals. The $p = \frac{1}{2}$ specialization is equivalent to Cauchy's additive equation, which admits wildly pathological non-affine solutions (Hamel, 1905) without some regularity hypothesis. Classical analysis from Cauchy (1821) to Ostrowski (1929) cataloged the minimal tameness conditions — continuity, measurability, boundedness on a set of positive measure — that rule the pathology out. These conditions became standard baggage in functional-equation arguments, carried reflexively even where the full continuous-coefficient equation is in play and no such baggage is needed.

This expository note makes the asymmetry explicit. We catalog which classical regularity hypotheses become vestigial under the continuous-coefficient equation, explain structurally why (the Hamel pathology lives at irrational p , exactly where the continuous-coefficient equation tests G), and then give a diagnostic that separates the saturated-Jensen settings in which these hypotheses are vestigial from the more common settings in which they remain load-bearing. The same five hypotheses appear on both sides of the line: in the axiomatic characterization of Shannon entropy they are indispensable, while in a calibration computation on an atomless space they are not, and we explain what distinguishes the two.

Keywords: Jensen equation, Cauchy equation, Hamel basis, functional equation, regularity hypothesis

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1. INTRODUCTION. One of the most serviceable inequalities in mathematics—Jensen's inequality, $\mathbb{E}[G(\xi)] \leq G(\mathbb{E}[\xi])$ for concave G —raises an immediate follow-up: when does equality hold for *every* two-point distribution supported on a given interval? The equation expressing universal two-point equality,

$$pG(u_1) + (1-p)G(u_2) = G(pu_1 + (1-p)u_2) \quad (u_1, u_2 \in I, p \in [0, 1]), \quad (\star)$$

imposed on a function $G: I \rightarrow \mathbb{R}$ on a real interval I , has at least three forms in classical analysis. These forms have been steadily conflated for over a century, with consequences for both the foundational theory and the modern applied literature where the equation surfaces.

Three forms of the equation. The *discrete-coefficient* Jensen equation,

$$G\left(\frac{u_1+u_2}{2}\right) = \frac{G(u_1)+G(u_2)}{2}, \quad (J_2)$$

is the $p = \frac{1}{2}$ specialization of $(\star)$. Setting $f(x) := G(x) - G(0)$, the equation (J_2) is equivalent on a translate of I to Cauchy's additive equation $f(x + y) = f(x) + f(y)$. Cauchy's equation inherits a full apparatus of pathological solutions: without measurability, monotonicity, boundedness on a set of positive Lebesgue measure, or another regularity hypothesis, the equation admits non-affine solutions constructed via a *Hamel basis*—a basis of $\mathbb{R}$ as a $\mathbb{Q}$ -vector space, whose existence requires the axiom of choice. The classical pathological-solution apparatus is the work of Hamel [1]; the regularity hypotheses sufficient to recover affineness were cataloged by Cauchy [2], Darboux [3], Sierpiński [4], Steinhaus [5], and Ostrowski [6].

The *rational-coefficient* Jensen equation,

$$pG(u_1) + (1-p)G(u_2) = G(pu_1 + (1-p)u_2) \\ (u_1, u_2 \in I, \quad p \in [0, 1] \cap \mathbb{Q}), \quad (J_{\mathbb{Q}})$$

strengthens (J_2) but does not escape its pathology. Cauchy additivity entails $\mathbb{Q}$ -homogeneity— $f(qx) = qf(x)$ for every $q \in \mathbb{Q}$ and $x \in \mathbb{R}$ —by an elementary induction on positive integers, sign reversal for negatives, and the inverse-of-multiplication argument for $1/n$. Combining additivity with $\mathbb{Q}$ -homogeneity returns $(J_{\mathbb{Q}})$ for f , hence for G . So $(J_{\mathbb{Q}})$, (J_2) , and Cauchy's equation share the same solution class up to constants, and all three inherit the Hamel-basis pathology.

The *continuous-coefficient* form $(\star)$, by contrast, is fundamentally different. The strengthening from $\mathbb{Q}$ to $\mathbb{R}$ in the range of p is *genuine*. As we will see, $(\star)$ does not admit any Hamel pathology: the pathology cannot survive at irrational p , and the continuous-coefficient equation tests G at every irrational. This is the subject of the paper.

The main result.

Theorem 1. *Let $M > 0$ and let $G: [0, M] \rightarrow \mathbb{R}$ satisfy $(\star)$ for every $u_1, u_2 \in [0, M]$ and every $p \in [0, 1]$. Then G is affine on $[0, M]$:*

$$G(v) = \frac{G(M) - G(0)}{M} v + G(0), \quad v \in [0, M].$$

No regularity hypothesis on G (continuity, measurability, monotonicity, boundedness) is required.

The proof is one line: a substitution into $(\star)$ that pins $G(v)$ on the straight line through the endpoints $(0, G(0))$ and $(M, G(M))$. We give it in Section 2 and use the remainder of the paper to articulate what it means.

The historical detour and the structural moral. Theorem 1 is elementary. The underlying substitution technique is folklore in functional equations: Aczél's [7, §2.1.4] contains a dyadic precursor proved (with a continuity hypothesis) by the same kind of substitution. The precise statement of Theorem 1 — continuous coefficients $p \in [0, 1]$, no regularity hypothesis on G — does not appear in this exact form in the standard references the author consulted; it is recorded here because the recurrence pattern of Section 5 needs it as a separable lemma. The contribution of the present paper is in three parts:

- *The explicit dictionary* (Section 3) of classical regularity hypotheses on G — continuity, measurability, monotonicity, boundedness on a set of positive measure,

and boundedness on the interval — each of which is required to force affineness for (J_2) but becomes *vestigial* under $(\star)$, together with the structural mechanism (the Hamel pathology lives at irrational p , exactly where $(\star)$'s continuous coefficient closes the door).

- *The strict-minimum hypothesis* (Section 4)—the equation $(\star)$ at a single endpoint configuration already suffices.
- *The diagnostic* (Section 5) for distinguishing saturated-Jensen derivations that genuinely produce the continuous-coefficient equation — where the dictionary's regularity hypotheses are vestigial — from those that produce a rational-coefficient relative, an additive function into a bounded codomain, or a Cauchy relative with a non-affine target. In the latter cases the same hypotheses are load-bearing; the misclassification runs in both directions, and we give examples of each.

Stylistically, this note follows the tradition of expository programs that concentrate on the part of Jensen-equation theory which does not require regularity assumptions on the function; Kuczma's [8] monograph is the canonical modern reference. The substitution that closes $(\star)$ is one of the simplest observations in that tradition; what justifies writing it down is the dictionary, the structural mechanism, and the survey of applied recurrences.

The classical detour through Hamel and the Polish school answered a perfectly natural question—*what regularity hypothesis suffices for $(J_2) \Rightarrow \text{affine}$?*—but a *different* question from the one that matters for many modern applications. The applied derivation does not produce (J_2) or $(J_{\mathbb{Q}})$; it produces $(\star)$ as a saturated identity, and the one-line substitution of Section 2 is the right closure tool. The dictionary of Section 3 makes explicit which classical regularity hypotheses become vestigial in this transition.

Terminology. Throughout the paper we refer to the substitution $u_1 = M$, $u_2 = 0$, $p = v/M$ in $(\star)$ — by which $G(v)$ is read off the straight line through the endpoints $(0, G(0))$ and $(M, G(M))$ — as the *endpoint substitution*. The substitution technique is standard folklore in the functional-equations literature (Aczél's [7, §2.1.4] proves a dyadic version); the descriptive label is our convenience, adopted for brevity in repeated reference rather than as a claim of established terminology.

2. THE RESULT AND ITS PROOF.

Proof of Theorem 1. Fix $v \in [0, M]$. Set

$$u_1 := M, \quad u_2 := 0, \quad p := \frac{v}{M}.$$

The constraints of $(\star)$ are satisfied: $u_1 = M \in [0, M]$, $u_2 = 0 \in [0, M]$, and $p = v/M \in [0, 1]$ since $v \in [0, M]$. Substituting into $(\star)$,

$$\frac{v}{M} G(M) + \left(1 - \frac{v}{M}\right) G(0) = G\left(\frac{v}{M} \cdot M + \left(1 - \frac{v}{M}\right) \cdot 0\right) = G(v).$$

Rearranging yields $G(v) = G(0) + (G(M) - G(0)) v/M$. ■

The endpoint substitution pins $G(v)$ on the straight line from $(0, G(0))$ to $(M, G(M))$ for every $v \in [0, M]$. Since this determines $G(v)$ at every v , the function G is fully determined by its values at the two endpoints — it is exactly the affine function joining them.

Corollary 2. *Under the hypotheses of Theorem 1, G is in particular continuous, monotone, locally Lipschitz, absolutely continuous, and Lebesgue measurable on $[0, M]$.*

Proof. Every affine function $G(v) = av + b$ on a bounded interval is differentiable with constant derivative $G' = a$, hence continuous; non-decreasing if $a \geq 0$ and non-increasing if $a \leq 0$; Lipschitz with constant $|a|$; absolutely continuous; and Borel-measurable, hence Lebesgue-measurable. ■

The order of inference matters and is worth stating explicitly. Under $(\star)$, the regularity properties of G are the *conclusion*, not the hypothesis. Each regularity hypothesis one might be tempted to assume—continuity, monotonicity, boundedness, measurability—is automatically satisfied by any solution of $(\star)$. The continuous-coefficient equation supplies its own regularity from within; no external regularity hypothesis is needed.

This is what makes the continuous-coefficient case structurally different from the discrete-coefficient case (J_2) , where some regularity hypothesis *is* required to rule out the Hamel pathology. We now examine the difference in detail.

3. THE DICTIONARY OF VESTIGIAL REGULARITY HYPOTHESES.

The dictionary. We place side-by-side the discrete-coefficient equation (J_2) and the continuous-coefficient equation $(\star)$, and ask which regularity hypotheses on G are *required* to force affineness for each. Table 1 lists five classical hypotheses, followed by a no-hypothesis row at the bottom. The first column names the hypothesis; the second records its status for (J_2) , with the standard historical attribution; the third records its status for $(\star)$. (For a modern textbook treatment of column 2—the regularity catalog for (J_2) —we refer the reader to Kuczma [8, §13.2].)

The bottom row is the point of the table. **No regularity hypothesis** suffices for (J_2) : there exists a non-affine, non-measurable, unbounded, additive function on $\mathbb{R}$ —the classical Hamel-basis pathology [1]—that supplies a non-affine solution of (J_2) . Conversely, **no regularity hypothesis** is *needed* for $(\star)$: Theorem 1’s endpoint substitution closes the proof at the level of the bare functional equation.

The five non-bottom rows tell a complementary story. For (J_2) , *some* hypothesis from these five is required; otherwise the Hamel pathology survives. For $(\star)$, *none* of these is required—and yet each is automatically satisfied by any solution (Corollary 2). Each hypothesis is, in the dictionary’s terminology, *vestigial*: classically required for (J_2) , classically expected for $(\star)$, but in fact unnecessary for $(\star)$.

The structural mechanism. Why does the Hamel pathology fail to apply to $(\star)$? The mechanism is direct and illuminating.

A Hamel-basis pathological solution G of Cauchy’s equation on $\mathbb{R}$ is $\mathbb{Q}$ -linear: $G(0) = 0$ and $G(qx) = qG(x)$ for every $q \in \mathbb{Q}$ and $x \in \mathbb{R}$. By design, G is *not* $\mathbb{R}$ -linear: there exists an irrational $r \in \mathbb{R}$ and an $x \in \mathbb{R}$ such that $G(rx) \neq rG(x)$.

The straight-line identity in $(\star)$ at the configuration $u_1 = M, u_2 = 0$ reads

$$pG(M) + (1-p)G(0) = G(pM). \quad (\star_0)$$

For a Hamel-pathological G (so $G(0) = 0$), $(\star_0)$ becomes $pG(M) = G(pM)$ —exactly the $\mathbb{R}$ -linearity assertion for G at the pair (M, p) . By $\mathbb{Q}$ -linearity of G , this holds for every $p \in \mathbb{Q}$. By the failure of $\mathbb{R}$ -linearity, it *fails* at some irrational p . So the Hamel-pathological G satisfies $(J_{\mathbb{Q}})$ (every rational coefficient is OK) but violates $(\star)$ at every irrational p where the $\mathbb{R}$ -linearity defect appears.

The structural point is one sentence:

Hypothesis on G	For $(J_2) \Rightarrow \text{affine?}$	For $(\star) \Rightarrow \text{affine?}$
Continuity on I: G is continuous at every $v \in I$.	Yes, suffices (Cauchy [2]).	No —it is the <i>conclusion</i> (Corollary 2).
Measurability on I: G is Lebesgue- or Borel-measurable on I .	Yes, suffices (Sierpiński [4]).	No —same.
Monotonicity on I: G is non-decreasing or non-increasing on I .	Yes, suffices (Darboux [3]).	No —same.
Boundedness on a set of positive measure: G is bounded on some Lebesgue-measurable $E \subseteq I$ with $ E > 0$.	Yes, suffices (Steinhaus [5]).	No —same.
Boundedness on I: G is bounded on all of I .	Yes, suffices (Ostrowski [6]).	No —same.
None: no regularity hypothesis at all.	Insufficient —Hamel-basis pathology [1].	Sufficient —Theorem 1.

Table 1. Regularity hypotheses on G . Column 2 records which hypotheses are classically required to force affineness for the discrete-coefficient equation (J_2) . Column 3 records their status under the continuous-coefficient equation $(\star)$.

The Hamel pathology lives at irrational p ; the continuous-coefficient equation $(\star)$ fore-closes on it there.

The classical program from Cauchy (1821) through Ostrowski (1929) can be read, in hindsight, as an investigation of what regularity hypothesis can substitute for the *unavailable* constraint at irrational p , once it became clear — after Hamel’s 1905 construction — that the rational-coefficient version of the equation does not by itself force affineness. With the continuous-coefficient version, the irrational-coefficient constraint is *available*, and the entire dictionary collapses.

4. VARIANTS AND LIMITS.

The strict-minimum hypothesis. The proof of Theorem 1 uses $(\star)$ at exactly one configuration: $u_1 = M$, $u_2 = 0$, with $p \in [0, 1]$ ranging freely. The full force of $(\star)$ —for every pair (u_1, u_2) —is not consumed. We record the strict minimum.

Theorem 3. Let $M > 0$ and let $G: [0, M] \rightarrow \mathbb{R}$ satisfy

$$p G(M) + (1 - p) G(0) = G(p M) \quad \text{for all } p \in [0, 1].$$

Then $G(v) = G(0) + (G(M) - G(0)) v/M$ on $[0, M]$.

Proof. Set $p := v/M$; then $p \in [0, 1]$ since $v \in [0, M]$. The hypothesis gives

$$p G(M) + (1 - p) G(0) = G(p M) = G(v),$$

so $G(v) = G(0) + p(G(M) - G(0)) = G(0) + (G(M) - G(0))v/M$. ■

Theorem 3 is *in principle* weaker than Theorem 1: the one-configuration hypothesis does not a priori imply the full $(\star)$. Under the conclusion (affineness) both hold; in practice an author who derives $(\star)$ from a richer setup (e.g., a Jensen-equality identity in a Bayes-risk computation; see Section 5) typically obtains $(\star)$ in full force. The narrower Theorem 3 is the right reference when one wants to know exactly how much of the equation the proof actually consumes.

Convex domains in higher dimensions. The endpoint substitution extends to any convex subset of a real vector space.

Theorem 4. *Let V be a real vector space, let $C \subseteq V$ be a convex set, and let $G: C \rightarrow \mathbb{R}$ satisfy*

$$pG(x_1) + (1-p)G(x_2) = G(px_1 + (1-p)x_2) \quad (x_1, x_2 \in C, p \in [0, 1]).$$

Then there is a linear functional $a: \text{span}(C - C) \rightarrow \mathbb{R}$ and a constant $b \in \mathbb{R}$ such that $G(x) = a(x - x_0) + b$ on C for any fixed $x_0 \in C$. Equivalently, G is affine on C .

Proof sketch. Apply Theorem 1 along every line segment in C : for each direction v with $x_0 + v \in C$, the map $\lambda \mapsto G(x_0 + \lambda v)$ is affine in $\lambda \in [0, 1]$. Hence $G(x_0 + v) - G(x_0)$ depends linearly on v in each direction; an inductive argument on the dimension of the affine span of the points involved promotes this to a global linear functional a on $\text{span}(C - C)$. The bookkeeping is standard and is given in [9]. ■

The structurally essential step is the one-dimensional endpoint substitution of Theorem 1. The higher-dimensional lifting is bookkeeping.

The rational-coefficient version retains the pathology. The strengthening from rational to continuous coefficients in $(\star)$ is doing genuine work: the rational-coefficient version $(J_{\mathbb{Q}})$ retains the Hamel pathology.

Proposition 5 (Folklore; cf. [8, §§5.2 and 13.2]). *There exists $G: [0, 1] \rightarrow \mathbb{R}$ satisfying $(J_{\mathbb{Q}})$ (hence (J_2)) on $[0, 1]$ that is not affine.*

Construction. Choose a Hamel basis H of $\mathbb{R}$ over $\mathbb{Q}$ containing 1; such a basis exists by Zorn's lemma applied to the partially ordered set of $\mathbb{Q}$ -linearly independent subsets of $\mathbb{R}$. Since H has cardinality $\mathfrak{c}$ and $\mathbb{Q} \cup \{1\}$ is countable, H contains an irrational $h \in (0, 1)$. Define a $\mathbb{Q}$ -linear map $\ell: \mathbb{R} \rightarrow \mathbb{R}$ by setting $\ell(1) = 0$, $\ell(h) = 1$, and extending $\mathbb{Q}$ -linearly on the remaining basis elements.

By construction, ℓ is $\mathbb{Q}$ -linear: $\ell(x + y) = \ell(x) + \ell(y)$ (Cauchy's equation) and $\ell(qx) = q\ell(x)$ for all $q \in \mathbb{Q}$ ($\mathbb{Q}$ -homogeneity). Set $G := \ell|_{[0, 1]}$; then G satisfies $(J_{\mathbb{Q}})$ on $[0, 1]$ because $\mathbb{Q}$ -rational convex combinations of points of $[0, 1]$ remain in $[0, 1]$ and ℓ respects $\mathbb{Q}$ -linear combinations.

Explicit witness that G is not affine: $G(h) = \ell(h) = 1 \neq 0$, while $G(q) = \ell(q) = q\ell(1) = 0$ for every rational $q \in [0, 1]$. Any affine map $A: [0, 1] \rightarrow \mathbb{R}$ agreeing with G on $\mathbb{Q} \cap [0, 1]$ would satisfy $A(0) = 0$ and $A(q) = 0$ for all rational q —forcing $A \equiv 0$, contradicting $G(h) = 1$. ■

Proposition 5 sharpens the dictionary: $(J_{\mathbb{Q}})$ on $[0, 1]$ admits non-affine solutions; only the strengthening to the continuous-coefficient $(\star)$ retires the regularity-hypothesis requirement.

5. TELLING $(\star)$ FROM ITS RELATIVES: A FIELD GUIDE TO SATURATED JENSEN. Theorem 1 closes the continuous-coefficient equation $(\star)$ with no regularity hypothesis on G ; Proposition 5 shows that the same closure fails for the rational-coefficient relative $(J_{\mathbb{Q}})$, where the Hamel pathology survives. A derivation in an applied field that pushes Jensen’s inequality to equality therefore lands on one side or the other of a sharp line, and which side it lands on decides whether a regularity hypothesis is doing real work. The purpose of this section is to draw that line precisely and to walk it. The conclusion is not that applied authors habitually carry unnecessary hypotheses; it is subtler and, we think, more useful: the saturated-Jensen identity arises from at least four structurally different sources, only one of which is the regularity-free situation of Theorem 1, and the four are easy to mistake for one another.

Four sources of a saturated identity, and one question that separates them. A saturated Jensen identity

$$p G(u_1) + (1 - p) G(u_2) = G(p u_1 + (1 - p) u_2)$$

is, before anything else, a *family* of assertions indexed by the weight p . Its mathematical content is precisely the set of weights for which the surrounding derivation actually licenses the assertion. Four sources recur.

The first is **finite mixing**. If the only primitive in the model is a binary mixture — combine two objects in some proportion — and compound objects are built by finitely iterating it, then the weights one can actually reach are the dyadic rationals, and, after closure under the operation, at most the rationals in $[0, 1]$. Such a derivation licenses $(J_{\mathbb{Q}})$, not $(\star)$; by Proposition 5 a regularity hypothesis is then genuinely required, and Theorem 1 does not apply.

The second is **additivity into a bounded codomain**. If the derivation delivers additivity of G together with the information that G takes values in a bounded set, then G is affine by the classical theorem that a bounded additive function is linear — the very results recorded in the last two rows of Table 1. Here a regularity hypothesis (boundedness) is load-bearing, but it is supplied for free by the codomain rather than assumed by hand, and again the operative mechanism is not the endpoint substitution.

The third is **a Cauchy relative with a non-affine target**. If the equation that actually appears is a relative of Cauchy’s equation whose intended solution is *not* affine, then a regularity hypothesis is needed to exclude Hamel-type pathological solutions, and $(\star)$ is simply not the equation in play. The characterization of Shannon entropy is the textbook instance (Section 5 below).

The fourth is **a genuine continuum of real weights**. If the weight p is a real parameter realized inside the structure — an atomless measure assigns to events every mass in $[0, 1]$, and Lyapunov’s convexity theorem [10] guarantees that every such mass is attained — and the saturated identity is asserted *directly* for that real p , with no detour through finite iteration, then the derivation licenses $(\star)$ at every real p , irrationals included. Only now does Theorem 1 apply, and only now is a regularity hypothesis on G vestigial.

This yields a single diagnostic question. *Does the derivation assert the identity at every real weight because the weights form a genuine continuum in the structure — or does it merely reach the rational weights, lean on the boundedness of the codomain, or solve a Cauchy relative whose answer is not affine?* In the first case the regularity hypotheses of Table 1 are vestigial and should be dropped; in the others they are load-bearing and must be kept.

The trap, then, is **misclassification**, and it runs in both directions. A derivation of one of the first three kinds can be mistaken for the fourth, so that an author drops a

hypothesis that was in fact indispensable. Conversely — and this is the rarer and more seductive error — a derivation of the fourth kind can be mistaken for a Cauchy problem, so that an author *adds* a defensive hypothesis that was never needed. The present paper exists because of an instance of the second error; we return to it in the closing note on provenance (Section 6).

Where the hypotheses are load-bearing. We begin on the side of the line where the classical hypotheses earn their keep — which, perhaps against first intuition, is where most of the familiar “saturated Jensen” derivations live.

Expected-utility representation (a continuum that is only apparent). In the von Neumann–Morgenstern tradition [11], sharpened by Herstein and Milnor [12], a preference relation over lotteries that is complete, transitive, independent, mixture-continuous, and Archimedean is represented by a utility functional that is linear in probability,

$$U(pL_1 + (1-p)L_2) = pU(L_1) + (1-p)U(L_2),$$

and these five conditions are jointly necessary and sufficient (see also [13, 14]). It is tempting to read the displayed identity as $(\star)$ and to offer Theorem 1 as a continuity-free substitute for the Archimedean axiom. The temptation should be resisted. The independence axiom is a condition on the *preference relation*; although it is stated for every real mixing weight, it does not by itself deliver a cardinal functional equation for a real-valued U . What produces a real-valued affine U in the first place is the **Archimedean axiom**, and it is genuinely indispensable: lexicographic preferences are complete, transitive, and independent, yet violate the Archimedean axiom and admit no real-valued representation whatsoever — so there is no U , and the displayed identity, and with it $(\star)$, is vacuous. Theorem 1 cannot replace the Archimedean axiom because Theorem 1 *presupposes* a real-valued solution of $(\star)$, and the Archimedean axiom is exactly what is needed to produce one. In the language of Section 5, the pre-Archimedean content is of the first kind; continuity is load-bearing.

The axiomatic characterization of Shannon entropy (a Cauchy relative). In the tradition of Khinchin [15] and Faddeev [16], the entropy $H(p_1, \dots, p_n) = -\sum_i p_i \log p_i$ is singled out by symmetry, a normalization, and a recursivity (grouping) axiom. The recursivity axiom does not reduce to $(\star)$. It reduces to the **fundamental equation of information**,

$$f(x) + (1-x)f\left(\frac{y}{1-x}\right) = f(y) + (1-y)f\left(\frac{x}{1-y}\right),$$

for the information function $f(x) = H(x, 1-x)$, and the intended solution is $f(x) = -x \log x - (1-x) \log(1-x)$, which is strictly *concave* — emphatically not affine. The fundamental equation of information is a relative of Cauchy’s equation, and like Cauchy’s equation it admits Hamel-type non-measurable solutions unless a regularity hypothesis excludes them [17]. This is the decisive point for the present paper, because the hypotheses that have been used to exclude them are *exactly the five rows of Table 1*. Faddeev assumed continuity of f outright [16] (a clean modern statement is in [18]); the requirement was then weakened, one classical hypothesis at a time — to integrability by Tverberg [19], to measurability by Lee [20], to boundedness on a set of positive measure by Diderrich [21], and to monotonicity on $[0, \frac{1}{2})$ by Kendall [22] and Borges [23]. The entropy characterization is thus the canonical place where the dictionary of Section 3 reappears in an applied setting — and there the hypotheses are *load-bearing*, because

the equation is a Cauchy relative with a non-affine target. It is the exact mirror of the calibration computation of Section 5, where a saturated identity of the same outward shape is genuinely $(\star)$; we record the contrast as a companion to Table 1 below.

Linear opinion pools (boundedness from the codomain). A third mechanism is worth naming briefly, because it too is easily mistaken for $(\star)$. If several experts' probability assignments are aggregated by a rule that commutes with marginalization, the rule must be a weighted average — a linear opinion pool — as soon as the underlying space has at least three points (McConway [24]; Aczél and Wagner [25]). The aggregation function is additive across disjoint events, and it takes values in $[0, 1]$ because its outputs are probabilities; linearity then follows from the bounded-additive theorem of Table 1, not from the endpoint substitution. As in the utility case, the saturated identity is real, but the work is done by a classical regularity property — here boundedness, supplied by the codomain — rather than by a continuum of weights.

Where the hypotheses are vestigial. Now the other side of the line, where the continuum of weights is real and Theorem 1 does its work.

Surrogate calibration on the resolution axis. Pass from the canonical loss-axis formulation of surrogate calibration [26] to a *resolution-axis* framing, in which risk is parametrized by the partition that aggregates the feature space rather than by the surrogate loss, and let $G(p)$ be the Bayes risk of a two-cell partition expressed in terms of the mass $p \in [0, 1]$ of one cell on an atomless probability space. A two-cell computation forces the saturated identity $(\star)$ on G : because the partition Bayes risk is, by construction, the mass-weighted average of the within-cell Bayes risks, the surrogate gap that Jensen's inequality otherwise leaves is forced to zero across the entire two-point family. Crucially, because the probability space is atomless the cell mass p ranges over *all* of $[0, 1]$ — every real value is the mass of some event, by Lyapunov's convexity theorem [10] — so the identity is tested at every real p , and Theorem 1 forces G affine with no continuity, measurability, monotonicity, or boundedness assumed. This is the seductive misclassification of Section 5 in the flesh: the saturated identity looks like a Cauchy problem and invites a defensive boundedness or measurability hypothesis on G , but the continuum of cell masses makes it genuine $(\star)$ and the hypothesis vestigial.

A self-contained toy. The mechanism is transparent in miniature, and the miniature also shows how thin the line is. Let $(\Omega, \mathcal{F}, \mu)$ be an atomless probability space and let $g: [m, M] \rightarrow \mathbb{R}$ be *any* function. Given values $u_1, u_2 \in [m, M]$ and an event A , let ξ be the two-valued random variable equal to u_1 on A and u_2 on its complement, and write $p = \mu(A)$. Because μ is atomless, as A ranges over $\mathcal{F}$ the mass p attains *every* value in $[0, 1]$ (Lyapunov's theorem again). Suppose g is mean-preserving across all such ξ — that the average value $\mathbb{E}[g(\xi)] = p g(u_1) + (1 - p) g(u_2)$ always equals the value at the average, $g(\mathbb{E}[\xi]) = g(p u_1 + (1 - p) u_2)$. This is exactly $(\star)$, now tested at every real p , so g is affine by Theorem 1, and nothing was assumed about its regularity. Replace the atomless space by a single fair coin — build the mixtures by tossing it repeatedly — and the attainable masses p collapse to the dyadic rationals; one is back in the first source of Section 5, Proposition 5 returns a non-affine solution, and a regularity hypothesis is once more required. The whole of Sections 3 and 5 is visible in the gap between an atomless measure and a coin.

The dictionary, mirrored. Table 1 records, for a *single* equation, which hypotheses force affineness. The diagnostic of this section is that the *same five hypotheses* sit on

opposite sides depending on the source of the identity. Table 2 places the load-bearing case of Section 5 (the entropy characterization, a Cauchy relative) beside the vestigial case of Section 5 (the atomless calibration computation, genuine $(\star)$); the contrast between its two columns is the content of the section.

Hypothesis on the unknown function	In the entropy characterization (a Cauchy relative)	Under a saturated $(\star)$ on an atomless space
Continuity	Required (Faddeev [16]).	Vestigial — a conclusion (Corollary 2).
Measurability	Required (Lee [20]).	Vestigial — same.
Monotonicity	Required (Kendall [22], Borges [23]).	Vestigial — same.
Boundedness on a set of positive measure	Required (Diderrich [21]).	Vestigial — same.
Boundedness on the interval	Required (a fortiori).	Vestigial — same.
None	Insufficient — Hamel-type solutions of the fundamental equation of information [17].	Sufficient — Theorem 1.

Table 2. The dictionary of Table 1, mirrored. The same five hypotheses are indispensable when the operative equation is the fundamental equation of information (a relative of Cauchy’s equation, whose solution is the concave Shannon function), and vestigial when a genuine continuum of weights makes the operative equation the continuous-coefficient $(\star)$.

The lesson. The recurrence worth articulating is not a habit of carrying unnecessary hypotheses; it is the ease of misreading which of the four sources one is in. Saturated Jensen is produced by finite mixing, by additivity into a bounded codomain, by Cauchy relatives with non-affine targets, and by genuine continua of real weights, and only the last retires the regularity hypotheses. The diagnostic of Section 5 is offered as the means of telling them apart; the mirrored dictionary of Section 5 is the warning that the same five classical hypotheses appear on both sides. An author who has internalized the Cauchy–Hamel toolkit is well defended against the first error and peculiarly exposed to the second — which is, as the closing note on provenance recounts, how this paper began.

6. CONCLUDING REMARKS. Theorem 1 is, mathematically, one line. The dictionary of Section 3 is a table. The structural argument of Section 3 is one sentence: *the Hamel pathology lives at irrational p ; the continuous-coefficient equation $(\star)$ forecloses on it there*. Why does this material deserve an article in the MONTHLY?

Two reasons. First, the classical detour through Hamel and the Polish school produced *foundational* results of classical analysis — and the dictionary of Section 3

articulates a precise sense in which the detour answered an adjacent question: not “what is the minimum hypothesis for $(\star) \Rightarrow \text{affine?}$ ” (Theorem 1’s answer: nothing), but “what is the minimum hypothesis for $(J_2) \Rightarrow \text{affine?}$ ” (Polish school’s answer: any of a dictionary of tameness conditions). Both questions are interesting; their answers are different.

Second, the diagnostic of Section 5 is, to our knowledge, not articulated explicitly elsewhere in the functional-equations literature. The same five regularity hypotheses turn out to be indispensable in some saturated-Jensen settings (the Shannon-entropy characterization) and vestigial in others (a calibration computation on an atomless space), and the explicit criterion may save other authors from misreading which case they are in — in either direction.

The endpoint substitution itself is the kind of one-line proof that, once seen, is hard to unsee — but that nonetheless takes some scaffolding to make visible. The present paper is the scaffolding.

A note on provenance. The present observation reached the author through an unconventional route: a *formalization project* on a separate topic, in which the lemma “ G satisfies $(\star)$ on $[0, M]$ implies G is affine” was initially stated with a defensive boundedness hypothesis on G , in deference to the Cauchy–Hamel literature. When the formal proof was written out, the boundedness hypothesis turned out to be unused: the endpoint substitution closed the proof on its own. A literature search then confirmed that the substitution technique is folklore: Aczél’s [7, §2.1.4] proves a dyadic precursor under a continuity hypothesis. The precise continuous-coefficient regularity-free statement of Theorem 1, while elementary, does not appear in this exact form in the standard references the author consulted; the present paper grew out of the attempt to explain, to the author’s own satisfaction, why the boundedness hypothesis had felt necessary in the first place, and to articulate the contrast between the discrete- and continuous-coefficient cases as a dictionary. The lesson is general: a defensive regularity hypothesis is not free, and a formalization project occasionally surfaces such hypotheses precisely because the formal system requires every clause of the statement to be used.

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